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# Head-Related Transfer Function

## Acquisition methods

Zweite Bachelorarbeit

Ausgeführt zum Zweck der Erlangung des akademischen Grades

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# Kurzfassung

In dieser Arbeit wird eine allgemeine Einführung zum binauralen Hören, kopfbezogenen Übertragsfunktionen (HRTFs) und Hoffnungen hinsichtlich personalisierter HRTFs gegeben. Weiters werden unterschiedliche Methoden zur Erstellung von HRTF Datensätzen dargelegt. Darüber hinaus wird ein Testsetting geschaffen, um künftig Methoden und deren resultierenden HRTFs zu evaluieren.

Kopfbezogene Übertragsfunktionen werden bereits verwendet, um über Kopfhörer eine Richtungswahrnehmung zu erzeugen, dabei wird Wissen über das binaurale Hören im Menschen zunute gemacht. Das Ziel der Virtualisierung von Audioinhalten ist es eine genaue Reproduktion des direktionalen Hörens, Raums und des Gesamterlebnisses zu bieten, welches eine Person im wirklichen Leben haben würde. Leider scheinen bestehende Systeme die HRTFs integrieren, diese Anforderungen derzeit nicht zu erfüllen. Die resultierende Lokalisierung kann unzureichend wirken und die Externalisierung unbefriedigend sein.

Aufgrund des komplexen Beschaffungsprozesses von HRTFs, verwenden die meisten Anwendungen derzeit eine oder manchmal mehrere vorgegebene Optionen von HRTF-Sets für einen Benutzer. Ein besserer oder schlechterer Virtualisierungseffekt wird mit einem nicht personalisierten HRTF-Set vermutet. Es gibt verschiedene Ansätze der HRTF-Personalisierung, welche das Ziel verfolgen erhöhte Genauigkeit der gesamten Virtualisierung zu liefern.

In dieser Arbeit wird versucht Klarheit zu gewinnen, inwieweit Virtualisierungen von Audioinhalten durch die Verwendung verschiedener Methoden zur Gewinnung von kopfbezogenen Übertragungsfunktionen und den daraus resultierenden HRTF-Sets beeinflusst werden. Eine Maßnahme zur Überprüfung der Qualität verschiedener Verfahren ist der Vergleich des Genauigkeitsgrades bei der Schallortung. Immersion und Externalisierung von Ton sind darüber hinaus Faktoren, welche die Erfahrung von virtualisierten Audioinhalten beeinflussen. Auch diese werden in dieser Untersuchung berücksichtigt.

In dieser Arbeit werden verschiedene Ansätze der HRTF-Individualisierung untersucht, und der Erwerb mancher umgesetzt. Für künftige Evaluierung der HRTF-Sets wird ein Testumfeld erstellt, welches die Platzierung virtueller Schallquellen ermöglicht und HRTFs in einem standardisierten Format laden kann. Zusätzlich werden Maßnahmen ergriffen, um Unregelmäßigkeiten der verwendeten Hardware auszugleichen und ein transparentes Ergebnis zu erzielen. Um zusätzlich Bewegung im Schallfeld zu ermöglichen wird Headtracking

integriert. Die gesammelte Erkenntnis wird analysiert und Schlussfolgerungen gezogen.

# Abstract

In this thesis a general introduction to binaural hearing, head related transfer functions, problems of HRTFs and hopes considering personalized HRTFs are given. Additional different methods of creating datasets of HRTFs are displayed. Furthermore, a test setting is created for future Evaluation of some methods with said integrated HRTFs.

Head-related transfer functions (HRTFs) are already in use to create a sense of direction through headphones, taking advantage of what is known about binaural hearing in humans. The goal of soundscape virtualization is to provide an accurate reproduction of directive hearing, room perception and an allover experience that a subject would have in real life.

Unfortunately, existing systems integrating HRTFs do not seem to meet these requirements at this point. The output of said can be inadequate regarding localization and unsatisfactory regarding externalization. Due to the complex acquiring process of HRTFs, most applications right now use one or sometimes a handful of fixed options of HRTF sets for a user. A better or worse virtualization effect using a non-personalised HRTF set is assumed.

Different approaches of HRTF personalization exist that aim to deliver an increased accuracy of the overall virtualization. This research is done to obtain an understanding, to what extent virtualizations of auditory scenes are influenced by using different acquiring techniques for head-related transfer functions and their resulting HRTF sets. A measure to check the quality of different methods is comparing the grade of accuracy in sound detection. Impacts like immersion and externalization of sound are other wanted features that influence the experienced performance in virtual soundscape applications. These are considered in this research as well.

Different approaches of HRTF individualization are investigated in this thesis, and the acquisition of some is practically executed. For future evaluation, a test setup is created that enables the placement of virtual sound sources and can load HRTFs in a standardized format. Further on movement in the sound field is enabled through head-tracking. Actions are taken to equalize irregularities of used hardware, providing a transparent outcoming.

The gained knowledge is analyzed and conclusions are drawn.

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# 1 Introduction

With the rise of binaural applications on headphones, the question can be raised what the ultimate goal of this development can be. A perfect representation of virtualized soundscapes as we experience them in the real world can be proposed. Such an application needs to provide an accurate reproduction of directive hearing, room and an allover convincing feeling a particular human has in real life.

This thesis investigates human binaural hearing and how this information is used to create virtualized soundscapes on headphones with the help of head-related transfer functions (HRTFs). Problems of existing applications are investigated and approaches that promise to deliver personalized HRTFs that improve the simulation are evaluated upon their method and effectiveness on localization. Other possible factors of improving soundfield virtualization are researched as well.

The goal of this thesis is to obtain an understanding, to what extent virtualizations of auditory scenes are influenced by using different acquiring techniques for head-related transfer functions and their resulting HRTF sets. For future comparison, a system that enables virtual sound source placement, HRTF set implementation and head tracking is set up. This setup shall provide a basis to compare and measure the localization errors of different HRTFs. Gained knowledge is analyzed and conclusions are made.

# 2 Binaural Hearing

Humans can generally localize sound sources in everyday situations to a certain degree even without the help of visual stimuli. Enabling to do so, binaural hearing, meaning hearing with two ears, is required for sound localization.

It was found that the brain uses differences in time and level information between the two ears to extract location information. Generally known as binaural cues the abbreviations of the terms interaural time difference (ITD) and interaural level difference (ILD) manifested. This is also known as the duplex theory. (Rayleigh, 1907)

## 2.1 Coordinate System

Usually, two types of coordinate systems are found when describing positions of sound sources relative to the head as seen in figure 1.

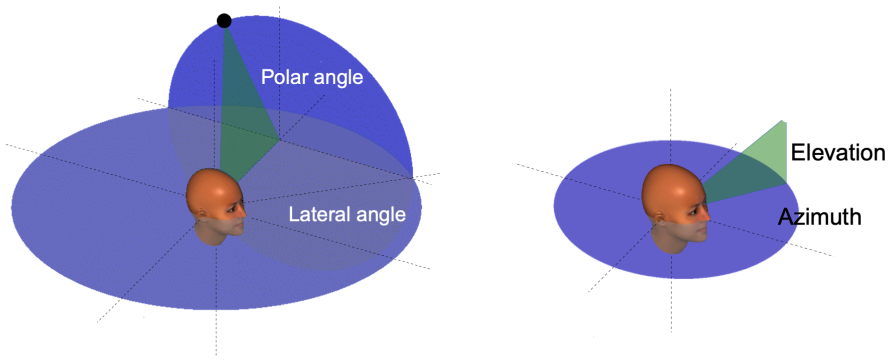


Figure 1: Interaural-polar coordinate (left); Spherical coordinate system (right) (Majdak, 2014)

A spherical coordinate system describes the horizontal direction through the azimuth angle and the vertical direction through the elevation angle. This system has the disadvantage that vertically placed poles do not correspond with the horizontally aligned ears of a listener but rather with the center of the head. (Figure 1, right)

An interaural-polar coordinate system (Figure 1, left) better represents the correspondence between the angles and the localization cues (Morimoto, 1984). In this system, the lateral angle represents the lateral direction of the sound and defines a sagittal plane parallel to the median plane (Figure 2). Within a sagittal plane, the polar angle (Figure 1, left) defines the direction of the sound source. (Majdak, 2014)

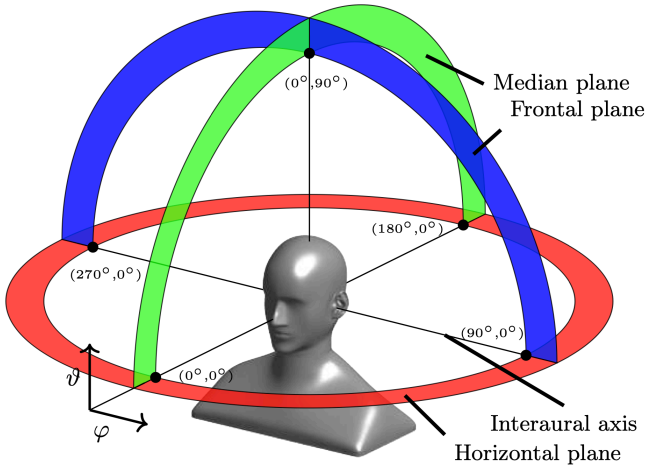


Figure 2: Interaural- polar coordinate system, describing planes

## 2.2 ITD

According to the duplex theory, different types of cues are utilized to determine a sound source. Low-frequencies were identified using interaural time differences, whereas high-frequencies are located using level differences. (Rayleigh, 1907)

It was later found that ITDs also can be extracted in high frequencies in the envelope of broadband signals with an upper limit of about 5000 Hz. The ITD groups are differentiated in the envelope ITD, meaning differences in the slowly-varying fluctuations in a signal and the fine-structure ITD, meaning the fast-varying fine structure differences in a signal. (Figure 3)

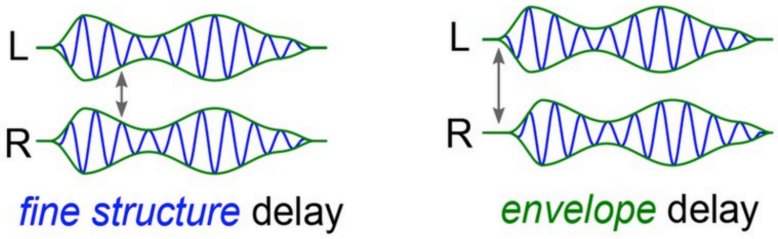


Figure 3: Interaural time differences (ITD)
(Stecker & Gallun, 2012)

ITDs are close to 0 µs for sound sources located in the median plane since they are equally distant from both ears. (Stecker & Gallun, 2012) Similar to this, each given value of ITD (or ILD) indicates a cone of symmetrical directions around the interaural axis rather than a single direction. This phenomenon is also known as “Cone of Confusion”. (Shinn-Cunningham et al., 2000)

## 2.3 ILD

In high frequencies, ILDs are more prominent. (Blauert, 1974) An interaural level difference results from sound being attenuated at the ear that is further away (head-shadow effect). The amount of ILD is frequency-dependent which means absorption, reflection and dampening of sound happens regarding head size and sound wavelength. An impression of distance in the nearfield is given as well, whereas the effect decreases further away. (Stecker & Gallun, 2012) The ILD can be between 20 to 30 dB for distances of 0,12 meter to 1 meter (Shinn-Cunningham et al., 2000).

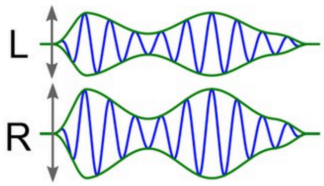


Figure 4: Interaural level differences (ILD)

## 2.4 Monoaural Cues

To distinguish a signal from the front or back, binaural hearing is of no big advantage. (Rayleigh, 1907) Directional characteristics of the body and most of all the pinna, known as spectral cues, play an important role not only in azimuth but also in elevated sound source detection. Asymmetries help maintain the sensitivity in elevation even in the median plane. Pinna cues are found to be more significant in higher frequencies. Investigations are ongoing to learn more about spectral characteristics nature. (Stecker & Gallun, 2012)

## 2.5 Head Movements

Other supporting cues resolving the cone of confusion problematic are head movements. Dynamic changes in ITD and ILD are produced when the head is moved, resulting in shifting an upfront source closer to one ear, having the opposite effect on the other ear for the same movement if the source is placed in the back. This effect can be utilized continuously and relatively subtle. (Stecker & Gallun, 2012)

## 2.6 Localization capability

It was found that the minimal audible angle (MMA), which presents the smallest detectable difference in angle of the same signal, is smaller for frontal sources and increases to the sides. Additionally, locability in noise was found to be generally better than in pure tones, where dips in certain specrral areas can be found. (Stecker & Gallun, 2012) The MAA for pure tones around the median plane goes down to a minimum of 1 degree azimuth for frequencies around 500 to 750 Hz and increases to a maximum of 3 to 4 degrees around 1.5 kHz and reaches a second minimum of around 2 degrees between 3 and 6 kHz. (Mills ,1958)

For elevation the MMA is fount to be larger varying from 2 to 4 degrees. (Perrott & Saberi, 1990)

# 3 Other psychoacoustic factors

To describe simulated or reproduced spatial audio quality of sound fields, some definitions and their aspects must be understood.

## 3.1 Immersion

The degree of immersion can be objectively assessed as the characteristics of a technology, for example, the addition of binaural cues to a headphone can technically be considered a step more immersive than a regular stereo headphone.

However, immersion can be seen as a more subjective measure for the feeling of oneself being submerged into, included, enveloped or surrounded by an environment. (Majdak, 2014) Approaches to measure this lead often to questions trying to empirically evaluate the felt grade of simulation, the involvement in terms of forgetting the real live environment involved, feeling oneself in the middle of the virtual environment and other. (Kailas & Tiwari, 2021)

## 3.2 Externalization

Sound externalization is the opposite of in-head localization and refers to the extent that a sound source can be located outside of a subjects head, a phenomenon that is often wished in virtualized sound scenes.

A common observation is that externalization is most robust when there is a great difference in ITD, leading to the effect that sound sources located close to the median plane are more likely to be internalized. (Leclere et al., 2019).

It was also found that natural binaural cues in ITD and ILD accelerate sound externalization, while manipulated cues lead to in-the-head localization. (Hartmann and Wittenberg, 1996) Additionally, pinna cues were found to be important especial for the poorly externalized midline locations. Further it was investigated that individualized spectral cues may not be critical for externalization. (Leclere et al., 2019)

Furthermore, if the direct sound contains only weak binaural cues (e.g., frontal sources), binaural cues from reflections seem to play a bigger role for externalization. (e.g., lateral sources). Overall reverberant sounds including up to 80ms of reverb tails are rather perceived externalized than anechoic sounds, with no further improvement for longer tails. Due to the natural higher absorption of high frequencies, they play a less important role in eternalization than low frequencies (Leclere et al., 2019).

Another significant impact to be mentioned is, that if signals are not found to be plausible and consistent regarding a subject's expectation about a room, also known as “room divergence effect”, externalization may break. (Klein et al., 2017). An increase of reported externalization was found, when visual information was given, like seeing the room that the source is presented or sound emerging from a source is animated. (Werner et al., 2016)

To sustain externalization, changes in location must be presented with no perceivable latency (Brimijoin et al., 2013).

The likelihood that a sound source appears externalized increases when the head rotates in relation to it, while self-initiated movements enhance externalization more than source movements. (Hendrickx et al., 2017)

## 3.3 Localization

The foundational localization cues are laid down in the chapter “binaural hearing”, some additional findings were made.

The term trading ratio describes the ability of a human to equalize between ITD and ILD to perceive a sound location in a flexible way. These ratios can range from less than 1 μs/dB to over 100 μs/dB and are dependent on spectral information, intensity, duration and reverberation. The resulting “images” of such experiments are distinguishable from accurate binaural scenarios. (Stecker & Gallun, 2012)

Sound source motion signals, e. g. a passing car, have been found to be hard to track, which is due to the fact that rapid change is not sensed by the binaural system well. The length of a signal motion is a key factor and requires a minimum stimulus duration of 150 to 300 ms. (Grantham, 1986)

Overall, ITDs and ILDs are sufficient to perceive the horizontal sound position accurately, while monaural and interaural spectral cues seem to have only a small impact. ITD was found to have a higher significance than ILD as a cue. (Macpherson and Middlebrooks, 2002)

# 4 Auditory Modeling Toolbox

With a focus on binaural models, the Auditory Modeling Toolkbox or AMToolbox is a Matlab/Octave toolbox for creating and using auditory perceptual models.

The AMToolbox was designed with the goal of enabling students and researchers to interact with and improve existing models. This goal is achieved by the consistent use of auditory models, comprehensive documentation, and user-friendly access. Publicly available human data, model demos, and model implementations are provided.

Additionally, model implementations can be assessed by conducting experiments that seek to replicate the findings from the relevant papers. It is accessible for free from „http://amtoolbox.sourceforge.net“. (Søndergaard & Majdak, 2013)

Some studies mentioned in this thesis are based on models available in the AMToolbox. To investigate human hearing it is of relevance.

# 5 Head-Related Transfer Function

Incoming sound is filtered due to reflection, shadowing, and diffraction by the ears, head, and torso before arriving at the listener's ear drums. This filtering changes with different positioning of the sound source. Containing the listener-specific directional acoustic features of one specific direction and the ITD, a pair of head-related transfer functions (HRTF) can describe one specific direction in relation to a subject's head. Whereas a set of hundreds of different binaural HRTF pairs from all directions is needed to describe the spatial features required for three-dimensional sound localization. (Wightman and Kistler, 1989).

In figure 5 a visualization example of head-related transfer functions of one ear is given.

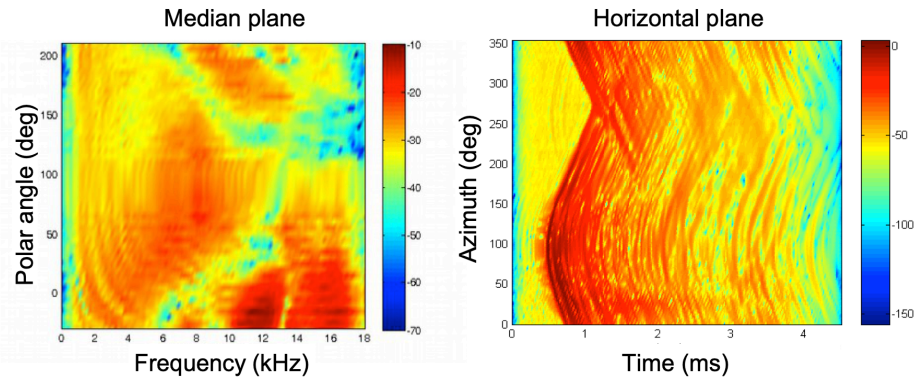


Figure 5: Left: Amplitude spectra of the directional transfer functions in the median plane. Right: Energy-time curves of the head-related impulse responses in the horizontal plane. The color encodes the amplitude in dB re max. (Madjak, 2014)

For distances larger than 1m HRTFs can be viewed as “far-field HRTFs”, below 1m they are sometimes called “near-field HRTFs”. When directional effects are under investigation, far-field HRTFs are assumed.

## 5.1 CTF and DTF

HRTFs may contain both direction-dependent and direction-independent features and thus can be thought of as a series of two acoustic filters. The direction-independent filter, represented by the common transfer function (CTF), can be calculated from an HRTF set comprising many directions by averaging the log-amplitude spectra of all available HRTFs of a listener's ear. The directional features are represented by the directional transfer functions (DTFs) which can be calculated by filtering the corresponding HRTF with the inverse CTF. The CTF usually exhibits a low-pass filter characteristic because higher frequencies are attenuated as an effect of the head and pinna shadow. Thus, compared to HRTFs, DTFs pronounce spectral features above 4 kHz and are commonly used to investigate the nature of spectral cues in localization experiments in sagittal planes (Majdak et al., 2010).

## 5.2 Application

Head-related transfer functions can be utilized in a variety of fields that have the goal to simulate natural hearing. This simulation is known under the term binaural synthesis, additionally virtual sound scapes as a whole system have become known as virtual auditory displays (VADs).

Some existing usage can be found in scientific research experiments, binaural auralization of room acoustics, virtual reality and communication. The most potential future uses for VADs will be in mobile communication and portable audio equipment, where improved speech intelligibility, enhanced sound orientation, and alleviated visual burden is promised.

Medical science can also use VADs to diagnose hearing problems. (Xie, 2013).

# 6 Personalized and non-personalized HRTF

In theory, the HRTFs and the proper headphone equalizations are all that is needed for a free-field simulation. Some early assumptions surrounding binaural synthesis proposed that it cannot deliver proper out-of-head localization or front-back distinction. A lack of attention to the fact that non-individualized HRTFs were used in studies that lead to these assumptions. Investigations that compare own binaural recordings, binaural recordings from other humans and binaural recordings from artificial heads showed statistically significant degradation when non-individual and artificial-head synthesis was utilized. These findings raised hope that personalized HRTF is able to produce a performance head to head to real life when properly applied. (Hammershøi & Møller, 2005)

## 6.1 Artificial Heads

While investigating artificial head performances on localization, it was found that the best performing artificial heads fall behind 60% of a variety of human heads. Comparing the HRTFs it can be seen that the spectral representation of artificial heads do generally not well represent the human HRTFs. Both investigations indicate that the differences between artificial heads and humans are larger than differences between humans. (Hammershøi & Møller, 2005)

## 6.2 Pinna-related notches

The spectrally lowest pinna-related notch is considered the most significant pinna-related spectral feature. Additionally, it was found that this notch is elevation-dependent, meaning it changes with different elevation angles. Generally, the pinna notch frequency shifts from an approximate range of 5 to 6kHz to a range of 12 to 13 kHz varying elevations from −40° to 60°. In the median plane, the notch increases with rising elevation and diminishes gradually at elevations greater than 60°.

With a standard deviation of 1050 Hz, no two or more of the 52 study subjects were found to show identical spectral features of HRTFs. Pointing out the significant intersubject variations in spectral locations and magnitude of these notches, a fraction of individuals may have shallow or even nonexistent pinna notches. Additionally, asymmetry in the details of pinna anatomy translates to asymmetric pinna notch frequencies in a subject.

While suspected that the pinna notch frequency may serve as a key localization cue, more work is needed to fully comprehend how the auditory system localizes, using this individual information. When non-individualized HRTFs are applied, the degradation of perceived performance in binaural applications may be explained by individualized pinna-related spectral properties. (Xie, 2013)

## 6.3 Directional spectral components

Seeking for spectral components that are responsible for the inter-individual differences in localization performance, a particular study decomposed HRTF sets at each location into average, lateral and intraconic spectral components along with an ITD. The theory behind this effort supports the idea that the individual components of an HRTF can be isolated from those that are more general in nature. By subtracting the directional spectrum from each location's matching lateral spectrum, the intraconic spectra are calculated. Individualized and non-individualized components were used to rebuild the HRTFs for a specific listener, and the modified HRTFs were then used to examine the effects of each alteration. According to the results, the cues crucial for HRTF individualization are nearly all present in the intraconic region of the HRTF spectrum, and the introduction of non-individualized signals into the other HRTF components barely affects localization. (Romigh & Simpson, 2014)

These findings claim to offer a base to understanding of which inter-individual variations in head-related transfer functions are most important for sound localization.

## 6.4 Effect of training

Substantial effects of auditory perception improvement involving training are known. Sound localisation is improved through selective attention and therefore,

in addition to the perceptual training, the enhanced attention to static cues like spectral profiles may explain the quick emerging benefits of such trainings. (Ahveninen et al., 2006).

An enhanced performance was achieved in different experiments. A particular study shows improved localization precision, and quadrant error (means the rate of front-back confusion). More precisely, quadrant errors were reduced largely for eye-level targets. The learning curve was flatter for elevated positions. After a short-term training period of 400 trials, subjects showed the biggest improvements, declining with subsequent training and not improving much more in relation to the initial success for training of up to 1600 trials.

In the study, subjects were tested in darkness, leading to the worst localization. When provided with a head-mounted display the localization performance improved and with added visual information on the target position while training, localization performance improved further. More precisely, naive subjects had lateral precision errors of 18 degrees and polar precision of 23.3 degrees. After the acoustic training, an improvement of lateral precision errors of 12.2 degrees and polar precision of 19.8 degrees was found. The provided stimulus was focused on learning the spectral cues by randomly roving the signal level in training. The spectral cues were those of listener-specific HRTFs. (Majdak et al., 2010)

of Using non-individualized HRTFs, to an extent accuracy improvements were found as well. A study undertaking perceptual training two 30-minute sessions with auditory, visual, and proprioceptive feedback shows improved localization accuracy in the front-back dimension only. (Zahorik et al., 2006)

While these studies are focused on short-term training, it has also been demonstrated that long-term training allows for the recalibration of the auditory system under visual guidance. Studies were able to change localization cues substantially, forcing a recalibration of the perceptual map. (Hyde & Knudsen, 2001).

# 7 SOFA Conventions and AES Standards

Due to the problem that head-related transfer function (HRTF) data has been stored in various formats in the past, initiatives were taken to create a system that allows interchangeable and transparent access to these datasets.

The Spatially Oriented Format for Acoustics (SOFA) was proposed and partually already in development as a data exchange format to solve a variety of issues and ultimately gives guidelines on how to store HRTFs, among other kinds of transfer functions (TF) and finite impulse response (FIR) filters, in a structured manner. (Majdak et al., 2013) This approach is continued to be developed in the AES69 standard, also known as SOFA 2.0, with its last official review 2020. (Majdak et al., 2020) To work with and edit the SOFA format it is necessary to understand some principles.

## 7.1 Format

SOFA relies on netCDF-4 to build upon, which is a set of software libraries and data formats that support creation, access, and sharing of scientific data. It is selfdescribing, networktransparent, machine independent, supports huge files, grants partial access within a file, and allows compression.

Information is stored in a single ".sofa" file through serializing the data inta a binary stream. (Majdak et al., 2020) (Unidata, NetCDF, 2022)

## 7.2 Data

When creating a SOFA file, a measurement setup is described by various objects and their relations. A measurement consists of data describing a single geometric condition and the resulting transfer function (TF). More precisely, this single TF is labeled with the corresponding dimensions and metadata. Information is stored in

a numeric container. All measurements combined are then stored in a single data structure.

While there exist predefined description conventions for the most common measurement setups, measurements are not limited to special cases like a fixed grid, or a constant distance. Potentially SOFA provides the flexibility to describe data of multiple conditions (listeners, distances, etc) in a single file. (Majdak et al., 2020)

## 7.3 Example

The AES69-2020 SOFA standard gives detailed information on describing data. Displayed is an example of a measurement setup considered in the SOFA convention as “SimpleFreeFieldHRTF”. Objects in SOFA consist of General, Receiver, Listener, Emitter, Source and Room. (Majdak et al., 2020)

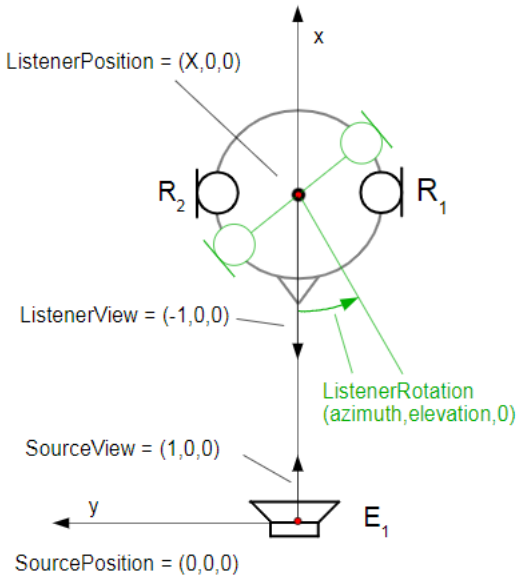


Figure 6: SimpleFreeFieldHRTF visualization (Majdak et al., 2020)

## 7.4 Application-programming Interfaces

While Matlab natively supports netCDF-4, other Application-programming interfaces (APIs) have been created as precompiled libraries for programming languages. (Majdak et al., 2020)

A list of 19 existing APIs for a variety of use cases and platforms can be accessed on “https://www.sofaconventions.org/mediawiki/index.php/Software_and_APIs”. Among them, applications for Matlab, Octave, C++, Python, JavaScript, VLC player and Cycling ´74s Max can be made of use.

## 7.5 Database

Databases holding HRTF sets of subjects can be accessed under “http://sofacoustics.org/data/database/”. They emerged for different initial use cases and institutions.

# 8 HRTF methods

Looking for a way to minimize errors in virtualizing binaural experiences, it was found that individualized HRTFs can have a major impact. Although much effort has been done over the years to suggest end-user-friendly solutions to HRTF customization, it still presents a challenge. Not all acquisition methods are accompanied by localization studies and if they are, there are wide variations in what geometrical range was tested, how many subjects were included, and what they were compared to. (Guezenoc & Seguier, 2018)

## 8.1 Categorization

The proposals addressing the matter can be categorized in acoustic measurement, numerical simulation, indirect individualization based on morphological data and indirect individualization based on perceptual feedback. (Guezenoc & Seguier, 2018)

## 8.2 Acoustic measurement

In a semi-anechoic or anechoic environment, loudspeakers are placed surrounding the subject from each direction of interest. Microphones in the ear capture the accompanying impulse responses to numerous positions. This process takes, depending on the methodology behind it, from over an hour to 4 minutes in modern approaches. Directional errors may occur since it is unlikely a subject can stay completely still throughout the whole measurement session. Flaws in the acoustic equipment or room can lead to errors as well. These errors can be avoided due to visual input and regulating mechanisms. Due to the rather static and complex setup, this method of producing a personalized HRTF is not a very consumer-friendly process. (Guezenoc & Seguier, 2018)

## 8.3 Numerical simulation

Simulating the dispersion of acoustic waves around a subject is also a common method for obtaining an individual HRTF set. The Fast-Multipole-accelerated Boundary Element Method (FM-BEM) for harmonic domain and the Finite Difference Time Domain (FDTD) for time domain have been the two main computing techniques. The Finite Element Method (FEM), raytracing and Differential Pressure Synthesis (DPS) have been explored in some works.

Acquisition of accurate geometry data is the key concern. A wide range of scanning techniques like molds, MRI, CT scan, etc. has been used. It was found that rearranging the subject's surface mesh ensures that the edge lengths are appropriate for the simulation. Remeshing resolution is a trade-off between numerical precision and processing time, higher numbers of mesh components increase computation time. Similar research has been done in FDTD simulation by examining the effects of the volumic geometry of a subject being voxelized on the resultant HRTFs. In general computation time can be reduced to a few hours nowadays.

In theory, this method is very end-user friendly, it only requires a 3D model of a listener, that could potentially be constructed utilizing photogrammetry using consumer-level smartphones. (Guezenoc & Seguier, 2018)

## 8.4 Indirect Individualiation based on morphological data

By modifying existing HRTFs, more listener specific HRTFs may be created. Such an individualization may be based on anthropometric data. Several researchers have investigated different methods of HRTF individualization within this context. (Guezenoc & Seguier, 2018) An example of such morphological categorization can be found in the CIPIC database as seen in figure 7 and 8. (Algazi et al., 2001)

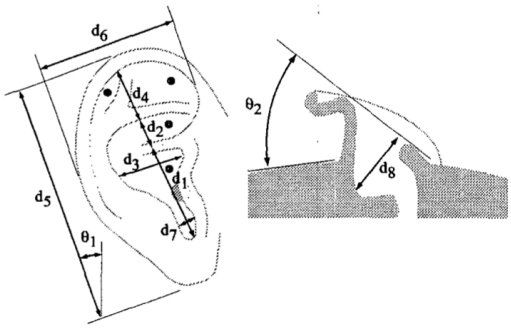


Figure 7: Pinna measurements as defined in the CIPIC database (Algazi et al., 2001)

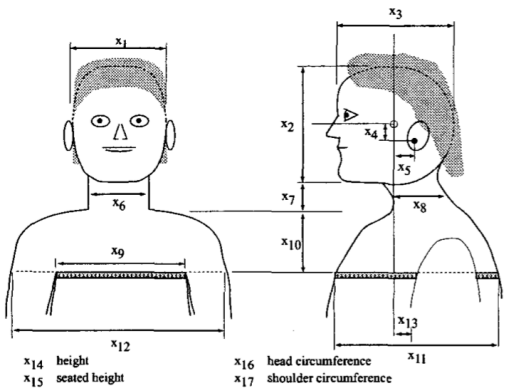


Figure 8 : Head and torso measurements as defined in the CIPIC database (Algazi et al., 2001)

By using the CIPIC database, a nearest neighbour approach that uses only 7 morphological parameters of a pinna picture, was evealuated and shows an average gain of 15% of elevation score. (Zotkin et al., 2002) Given that the perceptual study employed a custom perceptual score, it is hard to draw conclusions about these findings in contrast to others. (Guezenoc & Seguier, 2018)

Further on, another approach of utilizing such data is by using linear regression to calculate a scaling factor from measurements of the head and pinnae size. The method was evaluated to some extent upon the suggested method and the adaption of the generic HRTFs led to an increased perceptual performance compared to no individualization but a worse performance than achieved with a personal measured HRTF set. (Middlebrooks, 1999)

Additionally, an approach to predict a HRTF set by linear combination of HRTF sets using the coefficients of a model of anthropometric parameters was suggested but not perceptually evaluated. (Bilinski et al., 2014)

Other approaches of morphology based HRTF individualization acquisition implement statistical modeling with the help of Principal Component Analysis (PCA) or applicate machine learning mechanism like Independent Component Analysis (ICA). (Guezenoc & Seguier, 2018)

Surprisingly, none of these suggested approaches applied experiments in their research to evaluate the HRTF sets perceptual quality to an extent where conclusions regarding quality can be drawn. (Guezenoc & Seguier, 2018)

## 8.5 Indirect Individualization based on perceptual feedback

To suggest an alternative individualization technique, research suggests selection and adaption methods for HRTF sets based on a listener's response individual's based on the subjectively experienced spatialization.

A selection approach suggests a guided way of helping the listener to choose the best non-individual HRTF set among a database. A conducted localization experiment within the particular research showed better results than a non-individual HRTF sets but worse than a subject’s own set. (Seeber & Fastl, 2003) The experiment did not evaluate elevation perception in their research.

Additionally, non-induvidual HRTF set adaptation through perceptual feedback was researched in ways such as frequency scaling, filter- design-based tuning and statistical-model-based tuning.

Frequency scaling, as also mentioned at morphological individualization approaches, can be used for adapting a generic HRTF set. A companion study utilizing perceptual feedback in a form of a single tuning lever for the whole HRTF set is reported to gain a similar perceptual improvement in localization performance as the morphological approach within 20 minutes of adjustment. (Middlebrooks,

1999) Again, the achievable performance was found to be of less quality than personal measured HRTF sets. (Guezenoc & Seguier, 2018)

Filter-design-based approaches try to tune a generic HRTF set for a better result as well. The rather rough outcome of an easy implementation basically equalizes a whole HRTF set and is unpromising and to adapt a whole HRTF sets individual filters would be way too time-consuming. The output of such methods is not very well evaluated. (Guezenoc & Seguier, 2018)

Statistical-model-based tuning is suggested in order to minimize the number of tuning parameters with the help of models while retaining the capability to cover most of a database's HRTF space. PCA is the primary modeling technique that has been utilized.

A particular study suggesting fully perceptual-based individualization with an adaptive variational autoencoder promises a state-of-the-art method, relying on the modeling of HRTF sets with a neural network. The tuning procedure consisted in a gradient descent optimization of the network's weights where the cost was determined at every iteration by the user's notation of two HRTF sets presented to him by the algorithm. 18 out of 20 subjects had outcomes that were significantly better than an ideal match non-individual HRTF set. (Yamamoto & Igarashi, 2017) It is difficult to evaluate these results to those of other research due to the unconventional nature of the perceptual testing procedures. (Guezenoc & Seguier, 2018)

## 8.6 Examples

### 8.6.1 Acoustic Measurement

#### *8.6.1.1 Acoustic Research Institute, Inear-microphone measurement*

HRTFs are measured in a semianechoic chamber with 22 loudspeakers (figure 9) mounted at fixed elevations from -30° to 80°. In the center of the arc, the subject is seated, wearing in-ear microphones. A head tracker is employed to monitor head positioning and orientation during the measurement, leading to repeating a measurement if the subject moves beyond the acceptable range of 2,5 cm for positioning, 2,5° azimuth, and 5° elevation and roll. The measuring process takes about 20 minutes.

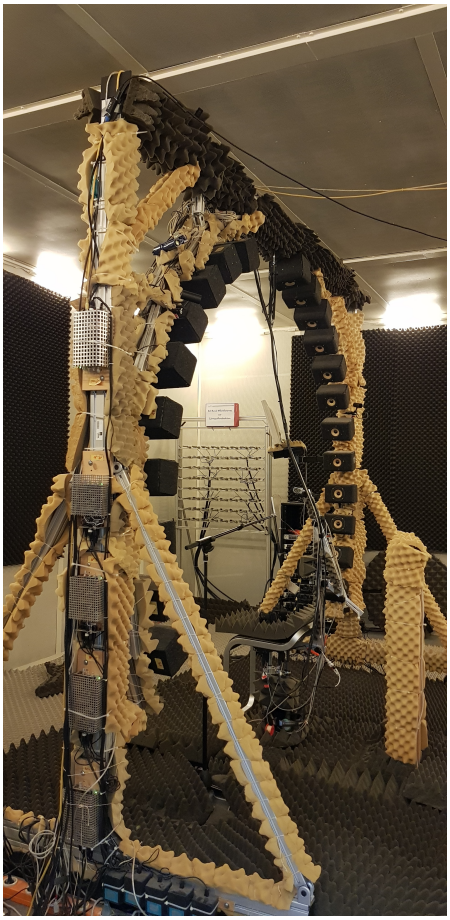

Figure 9: Loudspeaker array and motorized seat

Figure 10: Attached in-ear microphone

Figure 11: Subject on motorized seat performing the HRTF measurement

The HRTFs are measured for one azimuth and several elevations at once, using the multiple exponential sweep method (MESM), allowing interleaving and overlapping sweeps (50 Hz – 20 kHz) that are then recorded with the microphones.

Accordingly, the subject is rotated by 2,5° to measure HRTFs for the next azimuth. In the horizontal interaural plane, the HRTFs are measured with 2,5° spacing within the azimuth range of ±45°, and with 5° spacing outside this range. The positions of the HRTFs are distributed with a constant spherical angle, which means that the number of measured HRTFs in a horizontal plane decreased with increasing elevation, meaning at the elevation of 80°, only 18 HRTFs are measured. 1.550 HRTFs are measured in total for a subject using this setup.

It is assumed that the quality of HRTFs is vastly unaffected by the room and used equipment due to set measures. The equipment is covered with damping material to reduce the intensity of reflections, leading to a reverberation time of approximately 55 msec, and an amplitude of the greatest (floor) reflection 20 dB below the direct sound, delayed by at least 6.9 msec. The equipment transfer functions are measured through the in-ear microphones, placed in the center of the arc. The room impulse response is measured for all loudspeakers. The equipment transfer functions are removed from the HRTFs by spectral division. (Majdak et al., 2010)

### 8.6.2 Numerical Simulation

#### *8.6.2.1 MESH2HRTF*

As an open-source project MESH2HRTF aims to provide a software package for the numerical calculation of HRTFs. The application “Blender” is recommended to evaluate polygon meshes. “Openflipper” is recommended for preprocessing geometrical data. The process of MESH2HRTF consists of reading geometrical data of an object and calculating the corresponding sound field. As a result, an HRTF is put together and stored as a “.sofa” file. (Ziegelwanger, Kreuzer, et al., 2015)

In detail, sets of nodes form the geometric data into a 3D polygon mesh that describes a listener’s morphology. The calculation of HRTFs is schematically shown in figure 12. The head Ω with boundary Γ represents the scatterer. Ωe defines the domain outside the scatterer. A loudspeaker position at x∗ emits spherical waves. Γ∗ represents the microphone at the entrance of the ear canal. Among other mathematical steps to save computation time, the roles of sources and receivers are exchanged in the calculation. Meaning the in-ear microphones are simulated as a point source speaker. The pressure at however many loudspeaker positions is then evaluated at once and transformed into a HRTF set. (Ziegelwanger et al., 2016)

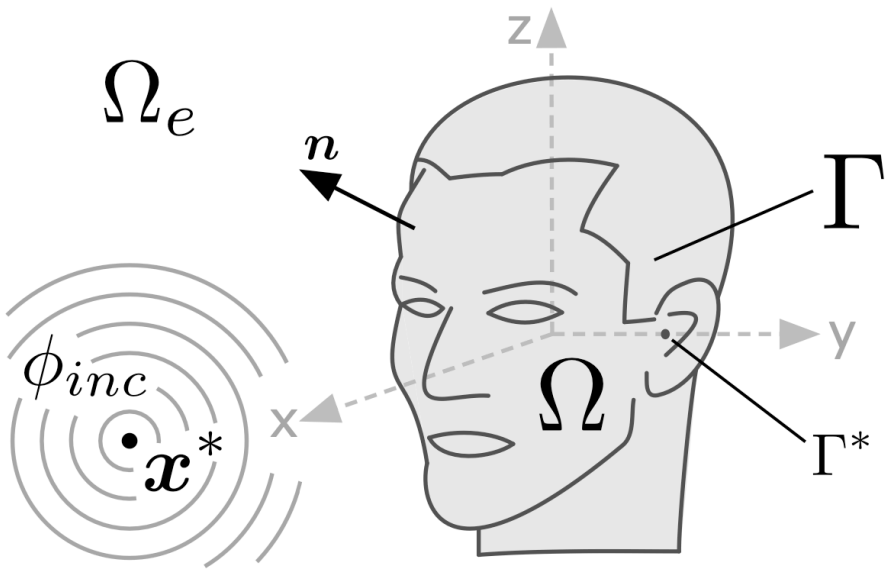


Figure 12: Schematic representation of the exterior scattering problem for the numerical calculation of HRTFs. (Ziegelwanger et al., 2015)

This acquisition method was evaluated, it was found that the mesh quality, concerning the resolution of the displayed listener's geometry, has an impact on the sound-localization performance in sagittal planes. The degradation of the geometrical detail of the pinna and to some extent also the head are the dominant factors, the torso was not considered in the evaluated calculation.

In particular, increasing the AEL (average edge length) of polygons leads to decreased performing HRTFs. It was found that AELs up to 2 mm provide similar virtual sound source localization performance as acoustic measured HRTFs. (Ziegelwanger, Majdak, et al., 2015) As seen in table 1 a modeled sagittal-plane localization experiment (Baumgartner et al., 2013) shows localization errors of virtual sound sources and compares acoustic measured and numeric calculated direct transfer functions (DTFs). A polar error of 29° and a quadrant error rate of 9.3 % is predicted. The base mesh model (1 mm AEL) predicted increases in the polar and quadrant error of 1° and 1.7 % and 2° and 2.7 % with 2mm AEL. Due to the small differences the outcoming is viewed as valid.

| Average Edge Length (mm) | | Number of Elements | Computation Time (hours) | Polar Error (°) | Quadrant Error (%) |
|---|---|---|---|---|---|
| Pinna | Head | | | | |
| Acoustic Measurement | | - | - | 29 | 9.3 |
| 0.4 | 0.5 | 2 488 720 | - | - | - |
| 1 | 2.5 | 111 456 | 19 | 30 | 11 |
| 2 | 2 | 87 706 | 14 | 32 | 12 |
| 3 | 3 | 44 190 | 8 | 33 | 14 |
| 4 | 4 | 25 442 | 6 | 38 | 17 |
| 5 | 5 | 16 464 | 4.5 | 40 | 17 |

Table 1: Mesh parameters and model localization performance.
(Ziegelwanger et al., 2013)

This is supported by the visual comparison of the calculated amplitude spectra to the acoustically measured data. In figure 13 a subjects left ears amplitude spectra for acoustically measured (AC) and calculated (BASE, 2 to 5 mm) DTFs are opposed. Notches, which are important features for locating sound sources in the median-plane, are shown by the dark regions in the amplitude spectra. Considering plausible explanations regarding some differences of AC DTFs and the calculated DTFs seem to be similar.

The small horizontal lines, which resemble reflections of the torso even though the torso is not part of the mesh, may occur as a result of the mesh being abruptly severed at the bottom of the neck without being smoothed. (Ziegelwanger et al., 2013)

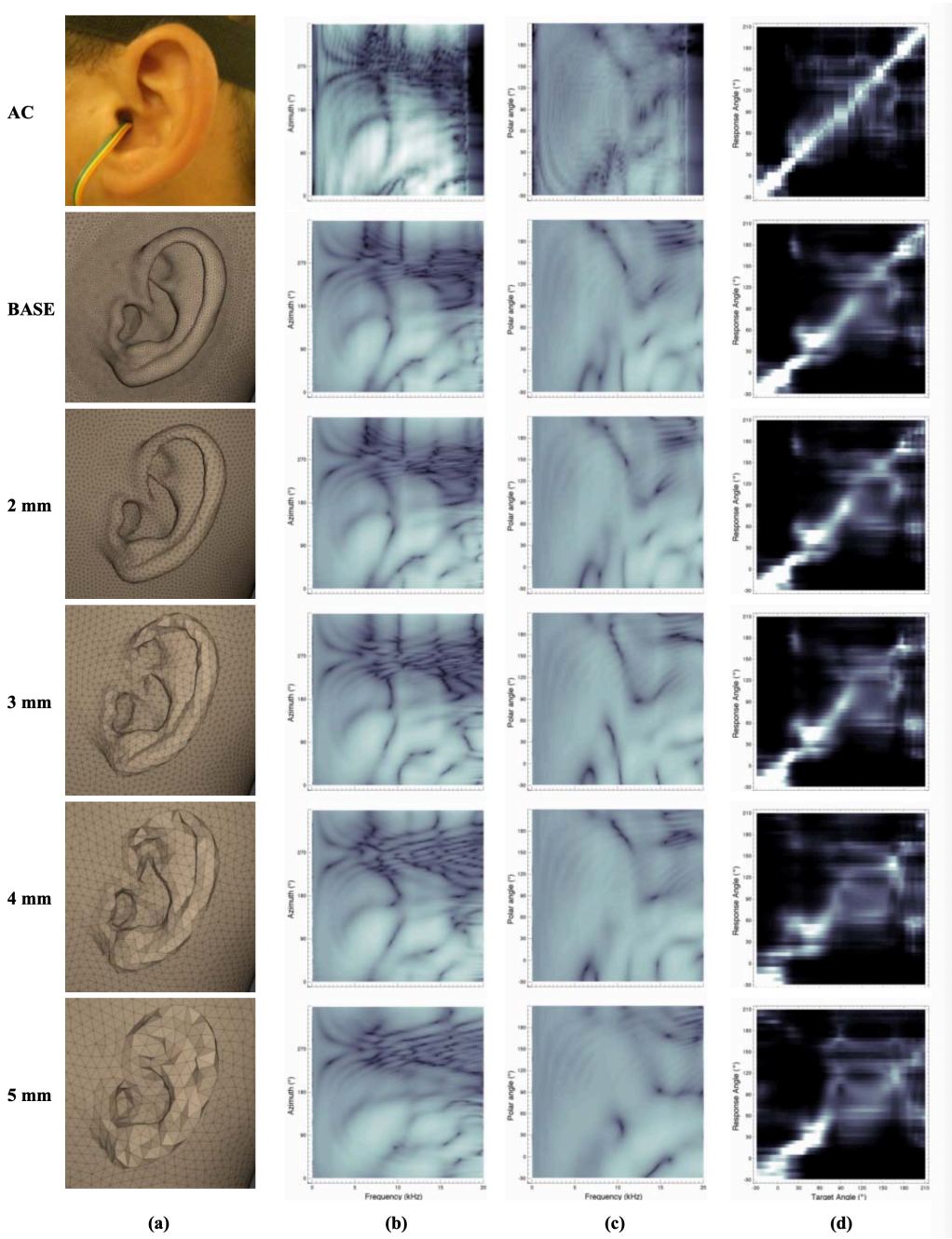


Figure 13: (a) pinna meshes; (b) DTF amplitude for the interaural horizontal plane (brightness: magnitude in dB); (c) DTF amplitude spectra for the median plane; (d) response probability matrix (brightness: response probability) (Ziegelwanger et al., 2013)

### 8.6.3 Individualization

#### *8.6.3.1 Fully Perceptual-Based 3D Spatial Sound Individualization with an Adaptive Variational AutoEncoder*

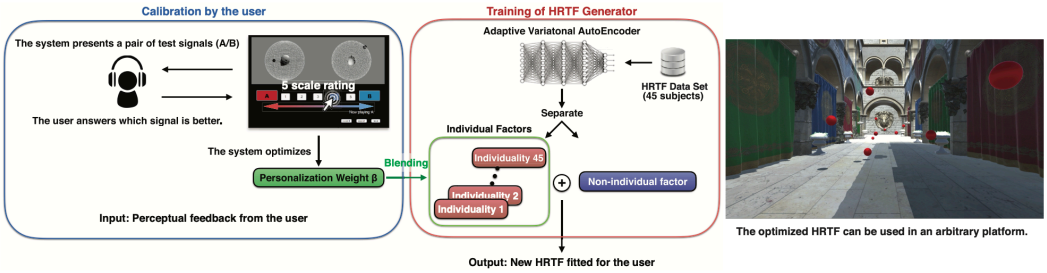


Figure 14: Schematic of perceptual-based HRTF individualization (Yamamoto & Igarashi, 2017)

The overall concept of this system is that a user can calibrate the head-related transfer function to his need based on his individual perception. This is done through continuous votes on two test signals, while each signal is applied with a different HRTF. An advantage of this system is, that it requires no additional equipment and the opstacles for a user are very low due to low efforts of obtaining this individual HRTF. The calibration process is claimed to take about 20 to 35 minutes. (Yamamoto & Igarashi, 2017)

The used algorithm applies an adaptive variational AutoEncoder (Rezende et al. 2014), which presents a form of machine learning, trained with the open available CIPIC database. (Algazi et al., 2001) The provided HRTFs in this database are decomposed for training the system. An adaptive variational AutoEncoder generates then individualized HRTFs for a subject during the perceptual feedback calibration by blending several HRTFs together. The algorithm is designed in a way, so it does not require optimizations for all the spherical directions around the head because the personalization weight is shared within all the directions. To cover all directions by running optimizations for only a few candidate directions claims to be a novelty in individualization approaches.

The method is evaluated by comparison with the best morphological matches in the CIPIC database and shows an improvement with their individualized HRTFs obtained for 20 subjects. A comparison to individual acoustically measured HRTF sets is not mentiones.

The authors claim that the system outputs the individualized HRTF in formats that are required by rendering platforms, if this includes the “.sofa” format is unclear.

# 9 Test setup

To create an environment for integrating HRTFs in a flexible and practical way the software Max, also known as Max/MSP/Jitter, was used to build upon. Max is a modular, real-time audio processing application. Through connecting objects, it can be used to develop interactive audio software, known as patches. Max allows third-party integration of libraries to extend the program with external objects. (Max, Software, 2022) The Spat 5 extensions were found to offer a good base for this test setup, containing more than 200 external objects focusing on sound spatialization, reverberation and multichannel activities. (Carpentier, 2019)

The patcher created for this setup utilizes the spat engines, specifically the “spat5.binaural” object, to render a binaural audio scene, presented over headphones. This setup is chosen upon its outstanding existing toolset, easy future expandability, and quick adaptability.

## 9.1 HRTF integration

The “spat5.sofa.loader” is used to integrate “.sofa” files in a variety of ways in the signal paths as seen in figure 15. The metadata in this file conveys, among other descriptive data, the HRTFs needed. The information is then read by the “spat5.binaural” engine and applied to the signal.

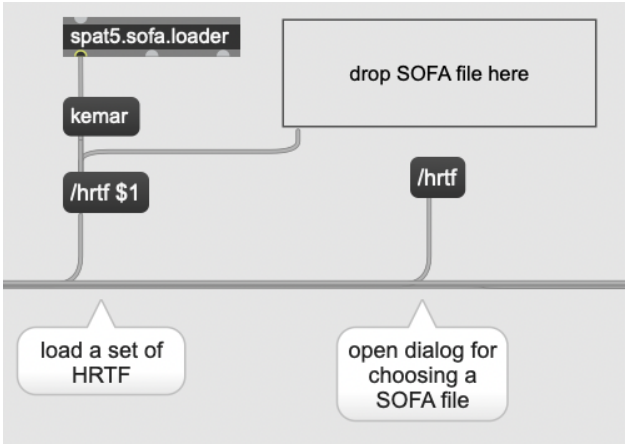


Figure 15: Patch section displaying “.sofa” file loading functionality.

## 9.2 Sources

Sources are initialized and can be set in a variety of formats (figure 16, left). Their positioning can be modified accessing the azimuth and elevation dials (figure 16, right). If needed the patch can be easily modified to enable dynamic sources additionally.

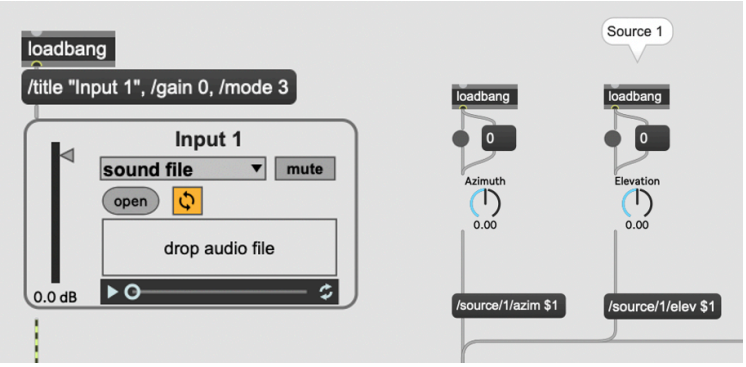


Figure 16: Patch section displaying source integration and manipulation functionality.

## 9.3 Head tracking

To capture motions of the head it is required to manipulate 3D geometrical data. Due to their non-interchangeability, different conventions are supported by the Spat 5 library. As seen in figure 17 quaternions and Euler angles can be utilized.

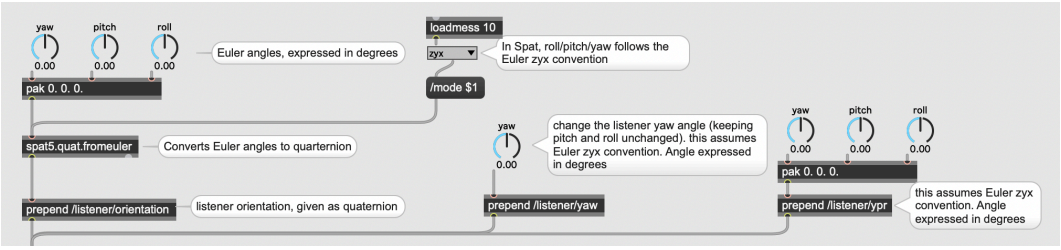


Figure 17: Subpatcher section displaying 3D geometrical data inplementation functionalities.

Generally, any device that can create the needed data, such as most modern smartphones can be integrated as a head-tracking device in this patch. As for existing solutions, the Waves Nx (*Waves Nx*, 2022) and Supperware (*Supperware*, 2022) head trackers were found and are available for this setup. They both can be easily attached to any headphone as seen in figure 18.

Figure 18: Left: Supperware head-tracker; Right: Waves Nx head-tracker

## 9.4 Headphone equalization

It is known that headphones do not have flat frequency responses that are required for authentic reproduction of binaural signals. This requires the transfer functions of the used headphones, particulary for scientific experiments, to be compensated for. Also one might consider that the headphone transfer function varies from the subject it is measured on. (Hammershøi & Møller, 2005)

Due to practicality the transfer function of the used headphones in this setup, a particular pair of “AKG 371s”, was measured five times on a “GRAS 45CA” test fixture with silicone ears, connected to an “APx515B Series” audio analyzer. The five resulting filters are averaged into one. This combined filter is integrated in the patch as a “.wav” file and convoluted with the signal. This was done due to the straightforward transformation and implementation. The filter can also be applied to the overall signal, achieving the same effect of equalizing the headphones irregular frequency response.

# 10 Acquisition methods evaluation

## 10.1 Measured

In the course of this research, measured HRTF sets of multiple subjects (16 in total) were able to be collected already. This was done with the described example setup at the acoustics research institute (ARI) in Vienna. The mainly automated process was conducted by ARI employees. The process took about 30 minutes up to an hour for most people. This differs because subjects must remain motionless during a measurement. If a subject fails to do so, measurements must be retaken, and adjusting to the target head angle was experienced to be difficult and time-consuming.

Overall, the experience feels straightforward but long, with the general disadvantage that setups like this are globally scarce, mainly used for research and therefore not open to the public. The approach is well established and irregularities that might occur are taken care of.

## 10.1 Calculated

The example methodology of a calculated HRTF acquisition for this thesis, “MESH2HRTF”, was explored as well. The biggest stepstone was the production of a 3D Mesh for subjects to be calculated. Different methods were found to be capable to deliver the required resolution and also representing the real-life object accurately. Photogrammetry with professional cameras as well as dedicated 3D scanning devices seem to yield the best results. (Reichinger et al., 2013) Whereas in the study photogrammetry approaches were rated worse with pictures taken on cellphones, nowadays this statement is outdated.

Due to the faster process (Reichinger et al., 2013) of capturing a geometry with dedicated 3D scanning devices, the initial goal for this research was to produce meshes of subjects heads through finding a service that specializes in 3D modelling with these devices. A sufficient test scan of such a service was only provided after already moving on to the fallback method, photogrammetry.

Multiple pictures with a state-of-the-art system camera were shot of in a spherical manner around 2 subjects' head. The process was conducted with a third-party

service which assembled the pictures into a 3D mesh with a dedicated photogrammetry software and applied some smoothing that is standard for such a process. The provided mesh fulfilled the requirements regarding the wished resolution of 1-2 mm AEL more than sufficient.

As seen in figure 19 hats were worn to provide a smoother outcoming in the photogrammetry process. It can be noticed the presence of hair (figure 19, left) led to a more distorted image compared to the real-life pinna within this trial.

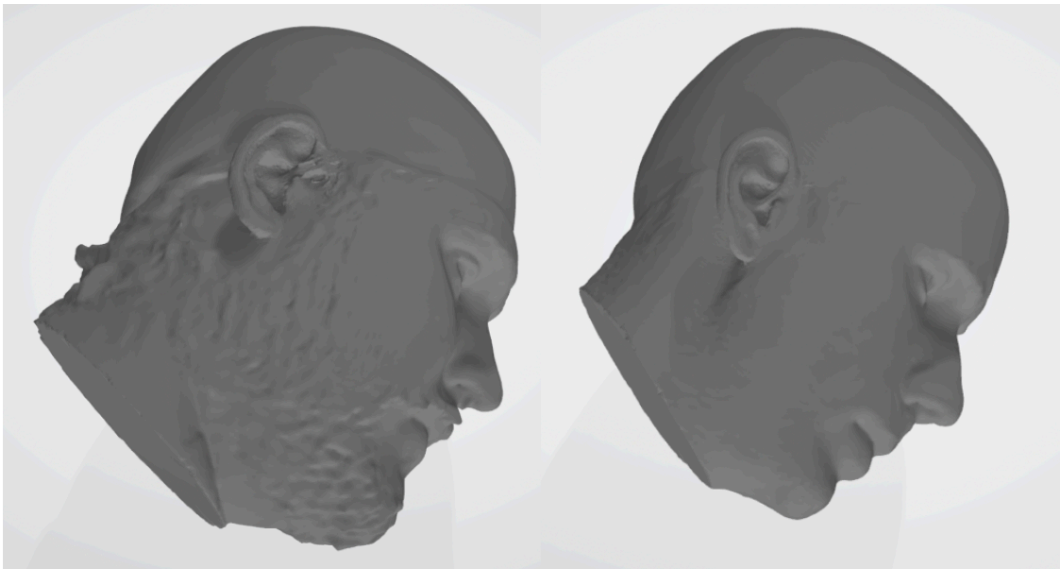

Figure 19: 3D model preview of scanned subjects

HRTF sets are now going to be calculated from the two meshes by the ARI utilizing the MESH2HRTF infrastructure. This is due to the fact, that even though the tool is open source, it is a challenging process to implement and perform the calculation. The resulting ".sofa" files can be investigated further and compared to the measured HRTFs in various ways.

Generally, this method, if done right, is quite promising because of the already validated perceptual outcoming. The main concern with this approach as of now is that the process is not streamlined. To model a subject's head as a 3D mesh sufficiently, the approach in its most efficient way seems to be at least as elaborate as measured HRTFs, especially because manual post-processing is needed. Technological advances in the near future might help to overcome these problems and help to push this approach forward.

For example, the possibility of using scanners in the form of slightly more advanced cellphones, equipped with lenses, depth sensors and most of all accompanying 3D modeling algorithms that do not need post-processing can potentially make this method available to a broad mass. Leading to fully personalized audio experiences that can be utilized by everyone without a need for special technology and is location independent.

The above described outlook was recently investigated in a study that compared the iPad Pro's LiDAR and TrueDepth capabilities to an industrial 3D scanning solution. The existing implementation in the iPad had higher deviations from the original, scanning was rated to be more difficult due to the location of the sensor and the scanning accuracy was defined to be not yet sufficient due to the low mesh resolution. The suggestion to give developers additional access and ongoing advancement of the sensor technology aligns with the progress wished for the "MESH2HRTF". (Vogt et al., 2021)

## 10.1 Individualization

Individualization approaches, both perceptual and based on morphological properties were not practically executed in the course of this research yet. A vast variety of different methods are suggested from different institutions, including the difficulty of lacking compareability in terms of location accuracy to well established methods. This makes it hard to make out a promising candidate to build upon.

While the suggested methods of individualizing mostly lead to improvement upon using a non-personalized HRTF set, within this research there was no evidence found for any method to get close to the quality of measure or calculation acquisition methods.

The reachable quality that can be achieved through individualization is unclear. A greater understanding of the interconnecting morphological features and HRTFs must be gained as well as an understanding if it is possible to tune whole HRTF sets with perceptual feedback for the result to be comparable with e. g. an individually measured HRTF set.

# 11 Conclusion

## 11.1 General

This research provides an understanding of human binaural hearing, accompanied by factors that may influence the quality of a virtual auditory display beyond head-related transfer functions alone. Head-related transfer function acquisition methods are categorized and found literature is evaluated in this research. Further promising acquisition methods were practically investigated upon their feasibility and evaluated. The gained experience is going to be the groundwork for future investigation, evaluation and comparison of acquisition methods and their resulting HRTF qualities and practicability.

With the existing and expandable test setup, acquired and to be acquired methods are going to be tested on localization quality, especially in the weak points that became proficient, for example in elevation and the cone of confusion regions.

## 11.2 Accessibility

The best available individualization approaches are without purpose if the indiduum that wants to utilize them cannot make use of them. This research shows that measured and simulation-based calculations with 3D scans are already evaluated and assumed to obtain the best performing HRTFs. Acoustic measurement setups lack a comfortable and accessible solution to be utilized by a broader mass of people. Simulation-based calculations build upon 3D meshes which are as of now comparably hard to obtain for consumers in a functional way. This results right now in the compromise of either offering no individualization at all or trying to accelerate the experience of virtually auditory systems to some state in between, utilizing individualization approaches.

Hopefully, in the upcoming years, the problems in 3D mesh creation will be solved. With big companies pushing the technology forward this could be a realistic outlook. On the other hand, an individualization solution that can reach the quality of measured or calculated approaches may emerge as well. Hybrid combinations of individualization approaches are a possible solution as well to provide a good experience for the user. Such a solution might be the selection of a set in a guided

way and tuning it then via perceptional feedback. A lot of combinations are possible but it is unknown as of now which can provide the best result.

## 11.3 Outlook on user applications

Once the technology is available, a natural, individualized auditory virtualization could find its integration into everyday life. For example, individualized VADs could be implemented as a standard in online meetings or recreational audio content like concerts could be attended with individualized virtualization in the future. More use cases can become the norm in the future.

## 11.4 Ongoing questions

Especially the unknown achievable quality of individualized HRTF sets and approaches is of interest and needs to be objectively evaluated. The most promising candidates of these methods need to be found and are of interest for ongoing investigations.

Further, the role of spectral features in sound localization on their own and their accordance with morphological features is of interest and needs more investigation.

An additional question that might also occur for future research is: In the overall implementation of a VAD, how relevant is the impact of a resulting HRTF set emerging of the presented methods upon all other factors such as head movement cues, visual and overall expectation, training, latency, etc., and vise versa. Also, not only localization but externalization and immersion need to be considered as well. Additionally, how to take the most efficient measures of accelerating these factors.

# Bibliography


Algazi, V. R., Duda, R. O., Thompson, D. M., & Avendano, C. (2001). The CIPIC HRTF database. *Proceedings of the 2001 IEEE Workshop on the Applications of Signal Processing to Audio and Acoustics (Cat. No.01TH8575)*, 99–102. https://doi.org/10.1109/ASPAA.2001.969552

Baumgartner, R., & Majdak, P. (2013). *Modeling sound localization of amplitude- panned phantom sources in sagittal planes*. 7.

Baumgartner, R., Majdak, P., & Laback, B. (2014). Modeling sound-source localization in sagittal planes for human listeners. *The Journal of the Acoustical Society of America*, *136*(2), 791–802. https://doi.org/10.1121/1.4887447

Bilinski, P., Ahrens, J., Thomas, M. R. P., Tashev, I. J., & Platt, J. C. (2014). HRTF magnitude synthesis via sparse representation of anthropometric features. *2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 4468–4472. https://doi.org/10.1109/ICASSP.2014.6854447

Blauert, J. (1997). *Spatial hearing: The psychophysics of human sound localization* (Rev. ed). MIT Press.

Carpentier, T. (2019). *A new implementation of Spat in Max*. 9.

Grantham, D. W. (1989). Motion aftereffects with horizontally moving sound sources in the free field. *Perception & Psychophysics*, *45*(2), 129–136. https://doi.org/10.3758/BF03208047

Guezenoc, C., & Seguier, R. (2018). HRTF Individualization: A Survey. *145th Audio Engineering Society Convention*. https://doi.org/10.17743/aesconv.2018.978-1-942220-25-1

Hammershøi, D., & Møller, H. (2005). Binaural Technique—Basic Methods for Recording, Synthesis, and Reproduction. In J. Blauert (Ed.), *Communication Acoustics* (pp. 223–254). Springer-Verlag. https://doi.org/10.1007/3-540-27437-5_9

Hartmann, W. M., & Wittenberg, A. (1996). On the externalization of sound images. *The Journal of the Acoustical Society of America*, *99*(6), 3678–3688. https://doi.org/10.1121/1.414965

Hendrickx, E., Stitt, P., Messonnier, J.-C., Lyzwa, J.-M., Katz, B., & de Boishéraud, C. (2017). Improvement of Externalization by Listener and Source Movement Using a "Binauralized" Microphone Array. *Journal of the Audio Engineering Society*, *65*(7/8), 589–599. https://doi.org/10.17743/jaes.2017.0018

Hyde, P. S., & Knudsen, E. I. (2001). A Topographic Instructive Signal Guides the Adjustment of the Auditory Space Map in the Optic Tectum. *The Journal of Neuroscience*, *21*(21), 8586–8593. https://doi.org/10.1523/JNEUROSCI.21-21-08586.2001

Kailas, G., & Tiwari, N. (2021). An Empirical Measurement Tool for Overall Listening Experience of Immersive Audio. *2021 IEEE International Conference on Consumer Electronics (ICCE)*, 1–5. https://doi.org/10.1109/ICCE50685.2021.9427770

Leclère, T., Lavandier, M., & Perrin, F. (2019). On the externalization of sound sources with headphones without reference to a real source. *The Journal of the Acoustical Society of America*, *146*(4), 2309–2320. https://doi.org/10.1121/1.5128325

Madjak, R., Madjak, & Dukes. (2020). *AES standard for file exchange—Spatial acoustic data file format*. 52.

Majdak, P. (2014). *Head-Related Transfer Functions and their Role in the Localization of Sound Sources in Human Listeners*. 44.

Majdak, P., Goupell, M. J., & Laback, B. (2010). 3-D localization of virtual sound sources: Effects of visual environment, pointing method, and training. *Attention, Perception, & Psychophysics*, *72*(2), 454–469. https://doi.org/10.3758/APP.72.2.454

*Max (software)*. (2022). https://en.wikipedia.org/wiki/Max_(software).

Middlebrooks, J. C. (1999a). Individual differences in external-ear transfer functions reduced by scaling in frequency. *The Journal of the Acoustical Society of America*, *106*(3), 1480–1492. https://doi.org/10.1121/1.427176

Middlebrooks, J. C. (1999b). Virtual localization improved by scaling nonindividualized external-ear transfer functions in frequency. *The Journal of the Acoustical Society of America*, *106*(3), 1493–1510. https://doi.org/10.1121/1.427147

Perrott, D. R., & Saberi, K. (1990). Minimum audible angle thresholds for sources varying in both elevation and azimuth. *The Journal of the Acoustical Society of America*, *87*(4), 1728–1731. https://doi.org/10.1121/1.399421

Rayleigh, Lord. (1907). XII. *On our perception of sound direction*. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, *13*(74), 214–232. https://doi.org/10.1080/14786440709463595

Reichinger, A., Majdak, P., Sablatnig, R., & Maierhofer, S. (2013). Evaluation of Methods for Optical 3-D Scanning of Human Pinnas. *2013 International Conference on 3D Vision*, 390–397. https://doi.org/10.1109/3DV.2013.58

Romigh, G. D., & Simpson, B. D. (2014). Do you hear where I hear?: Isolating the individualized sound localization cues. *Frontiers in Neuroscience*, *8*. https://doi.org/10.3389/fnins.2014.00370

Seeber, B. U., & Fastl, H. (2003). *SUBJECTIVE SELECTION OF NON-INDIVIDUAL HEAD-RELATED TRANSFER FUNCTIONS*. 4.

Shinn-Cunningham, B. (2000). Adapting to Remapped Auditory Localization Cues: A Decision-Theory Model. *Perception and Psychophysics*, 42.

Søndergaard, P. L., & Majdak, P. (2013). The Auditory Modeling Toolbox. In J. Blauert (Ed.), *The Technology of Binaural Listening* (pp. 33–56). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-37762-4_2

Stecker, G. C., & Gallun, F. J. (2012). *Binaural Hearing, Sound Localization, and Spatial Hearing*. 53.

*Supperware*. (2022). https://supperware.co.uk/headtracker-overview

*Unidata, NetCDF*. (2022, May). https://www.unidata.ucar.edu/software/netcdf/

Vogt, M., Emmelmann, & Rips. (2021).

*Waves Nx*. (2022). https://www.waves.com/nx

Werner, S., Klein, F., Mayenfels, T., & Brandenburg, K. (2016). A summary on acoustic room divergence and its effect on externalization of auditory events. *2016 Eighth International Conference on Quality of Multimedia Experience (QoMEX)*, 1–6. https://doi.org/10.1109/QoMEX.2016.7498973

Wightman, F. L., & Kistler, D. J. (1989). Headphone simulation of free-field listening. I: Stimulus synthesis. *The Journal of the Acoustical Society of America*, *85*(2), 858–867. https://doi.org/10.1121/1.397557

Xie, B. (2013). *Head-related transfer function and virtual auditory display* (Second edition). J. Ross Publishing.

Yamamoto, K., & Igarashi, T. (2017). Fully perceptual-based 3D spatial sound individualization with an adaptive variational autoencoder. *ACM Transactions on Graphics*, *36*(6), 1–13. https://doi.org/10.1145/3130800.3130838

Zahorik, P., Bangayan, P., Sundareswaran, V., Wang, K., & Tam, C. (2006). Perceptual recalibration in human sound localization: Learning to remediate front-back reversals. *The Journal of the Acoustical Society of America*, *120*(1), 343–359. https://doi.org/10.1121/1.2208429

Ziegelwanger, H., Kreuzer, W., & Majdak, P. (2015). *MESH2HRTF: AN OPEN-SOURCE SOFTWARE PACKAGE FOR THE NUMERICAL CALCULATION OF HEAD-RELATED TRANFER FUNCTIONS*. https://doi.org/10.13140/RG.2.1.1707.1128

Ziegelwanger, H., Kreuzer, W., & Majdak, P. (2016). A priori mesh grading for the numerical calculation of the head-related transfer functions. *Applied Acoustics*, *114*, 99–110. https://doi.org/10.1016/j.apacoust.2016.07.005

Ziegelwanger, H., Majdak, P., & Reichinger, A. (2013). Calculation of listener-specific head-related transfer functions: Effect of mesh quality. *The Journal of the Acoustical Society of America*, *133*(5), 3282–3282. https://doi.org/10.1121/1.4805374

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